Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 1 (Answer Key)

Date: _____ Name: **TEACHER ANSWER KEY**

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**30**
Part 1	**25**
CW	**5**
Marks Obtained	**25 / 25**

Section A:

MCQ: **(5x1=5 marks)**

For each question 1-5, four options are given. Circle the correct option i, ii, iii or iv.

1. Which number system uses Base-16 and the digits 0–9 along with letters A–F?

i. Binary
ii. Denary
iii. **Hexadecimal**
iv. Octal

2. How many bits make up exactly 1 Byte of digital data?

i. 4 bits
ii. **8 bits**
iii. 16 bits
iv. 1,024 bits

3. How many bits are used by the standard ASCII character set to represent one character?

i. **7 bits**
ii. 16 bits
iii. 32 bits
iv. 64 bits

4. What is the standard unit used to measure the Sample Rate of a digital sound wave?

i. Pixels per inch (PPI)
ii. Bytes per second
iii. **Hertz (Hz)**
iv. Bits per pixel

5. Which type of file compression reduces file size WITHOUT losing any original data?

i. Lossy compression
ii. **Lossless compression**
iii. Analogue compression
iv. Pixel compression

Section B:

True or False: **(5x1=5 marks)**

6. Write True or False in the box provided for each of the following statements:

i.	Computers use the Binary (Base-2) number system because digital switches only have two states: ON (1) and OFF (0).	**True**
ii.	An image with an 8-bit colour depth can display a maximum of 256 different colours.	**True**
iii.	Decreasing the sample rate and sample resolution of an audio file will make the sound quality better.	**False**
iv.	Lossy compression permanently removes unnecessary data that humans cannot easily notice.	**True**
v.	In binary storage units (IEC standard), 1 Kibibyte (1 KiB) is equal to 1,000 Bytes.	**False**

Section C:

Fill in the Blanks: **(5x1=5 marks)**

7. Fill in the blanks using the correct words from the box below:

Word Bank: [Pixel | ADC | Unicode | MAC Address | Metadata]

i. A **Pixel** is the smallest single dot or picture element that makes up a bitmap image.

ii. The **Unicode** character set uses 16 to 32 bits to represent characters from all world languages and emojis.

iii. An **ADC** (Analogue-to-Digital Converter) converts analogue sound waves into digital binary data.

iv. Image **Metadata** stores extra file information such as image dimensions (width and height) and colour depth.

v. A **MAC Address** is a unique hardware identifier written in Hexadecimal (for example, 00-1C-B3-4F-25-FE).

Section D:

Number Systems & Hexadecimal: **(3+2=5 marks)**

8. Complete the following number conversions and write the final answer in the table:

No.	Conversion Question	Final Answer
i.	Convert 8-bit Binary **1010 1110** to Denary (Base-10): **Working: 128 + 32 + 8 + 4 + 2 = 174**	**174**
ii.	Convert 12-bit Binary **0111 1101 0100** to Hexadecimal: **Working: 0111 = 7, 1101 = 13 (D), 0100 = 4**	**7D4**
iii.	Convert Hexadecimal **5B8** to a 12-bit Binary number: **Working: 5 = 0101, B (11) = 1011, 8 = 1000**	**0101 1011 1000**

9. State two reasons why programmers prefer using Hexadecimal instead of long Binary strings: (2 marks)

i. It is much shorter and easier for humans to read, write, and remember than long binary strings.

ii. Fewer mistakes are made when copying numbers, and it is very easy to convert to/from binary.

Section E:

Data Compression & File Size: **(3+1+1=5 marks)**

10. Tick (✓) the most suitable compression method for each file below, and answer (iv) & (v):

No.	File Type / Scenario	Lossy (✓)	Lossless (✓)
i.	A photograph saved as a **JPEG** file for a school website.	✓	
ii.	A computer software program file (**program.exe**) or source code.		✓
iii.	An **MP3** music file used for online audio streaming.	✓	

iv. Compress this pixel row using **Run-Length Encoding (RLE)**: **W W W W W W B B B W W W W B B B**

RLE Answer: **6W 3B 4W 3B** **(1 mark)**

v. A plain text file has **4,096 characters** (1 Byte each). Express its file size in **KiB** (1 KiB = 1,024 Bytes):

File Size: **4,096 / 1,024 = 4 KiB** **(1 mark)**